Sea Level Rise Scenario Tool

ArcGIS ModelBuilder Reproduction - Unit 07

This report documents the traditional ArcGIS ModelBuilder assignment requirements and a Python reproduction of the same processing logic using the course-provided DEM. The actual course submission should include the ArcGIS toolbox file and screenshots of the ModelBuilder canvas and tool interface.

Item	Value
Input raster	unit07-010-n32_w081_1arc_v2.tif
CRS	EPSG:4326
Raster size	3601 x 3601 cells
Elevation range	-88.0 to 105.0 meters
Default sea-level change	6 feet
Default area-of-concern radius	500 feet

Assignment Requirement Checklist

Requirement	How the model satisfies it
Input raster parameter	The tool prompts for an elevation raster dataset in meters.
Sea-level change parameter	The user supplies an integer threshold in feet.
Area-of-concern radius parameter	The user supplies a buffer distance in feet.
Output polygon path parameter	The user chooses the final polygon output path.
Raster Calculator 1	DEM values are converted from meters to feet.
Raster Calculator 2	A 0/1 inundation raster is generated from the sea-level threshold.
Raster to Polygon	The 0/1 raster is converted into polygon features.
Make Feature Layer	Only submerged polygons are selected.
Buffer	The selected polygons are buffered to create the final area of concern.

ModelBuilder Workflow

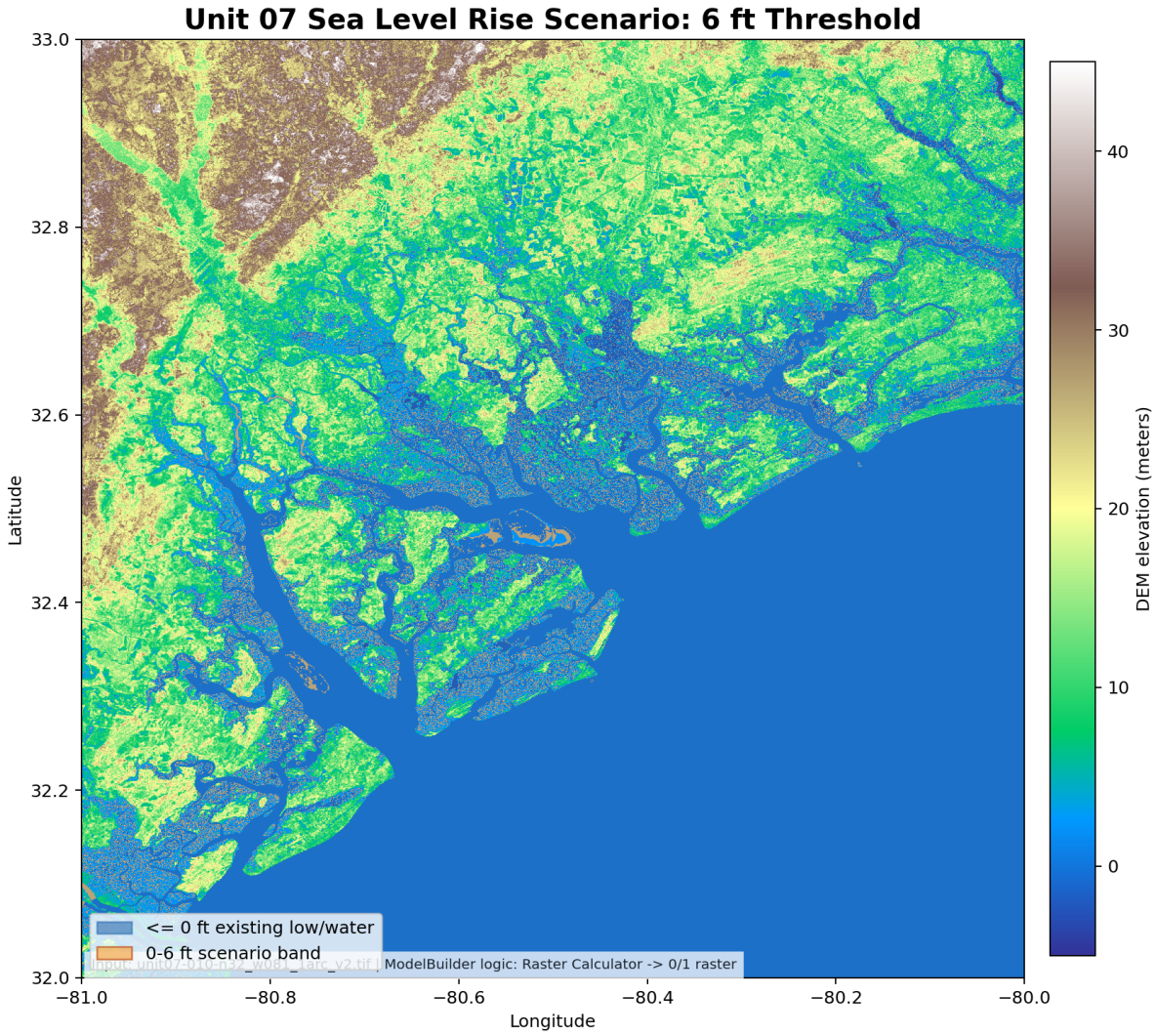
ArcGIS ModelBuilder Unit 07 Required Workflow

Default Testing Scenario Summary

Metric	Value	Unit	Note
Input DEM	unit07-010-n32_w081_1arc_v2.tif	file	Course-provided 1 arc-second DEM raster.
Raster size	3601 x 3601	cells	Full raster grid used for threshold and polygonization.
Sea level change default	6	feet	Default user-facing tool parameter.
Area of concern radius default	500	feet	Default user-facing buffer parameter.
Valid DEM cells	12967201	cells	Cells with usable elevation values.
Inundated cells	6027770	cells	Cells where elevation in feet is less than or equal to the sea-level parameter.
New scenario band cells	458956	cells	Cells above current sea level and within the default rise threshold.
Inundated polygon count	20996	features	Raster-to-polygon output before dissolve.
Inundated area	4850.14	sq km	Projected area in UTM Zone 17N.
New scenario band area	368.93	sq km	Area newly highlighted between 0 ft and the sea-level parameter.
Final area of concern	6402.20	sq km	Dissolved inundation polygon buffered by the default radius.

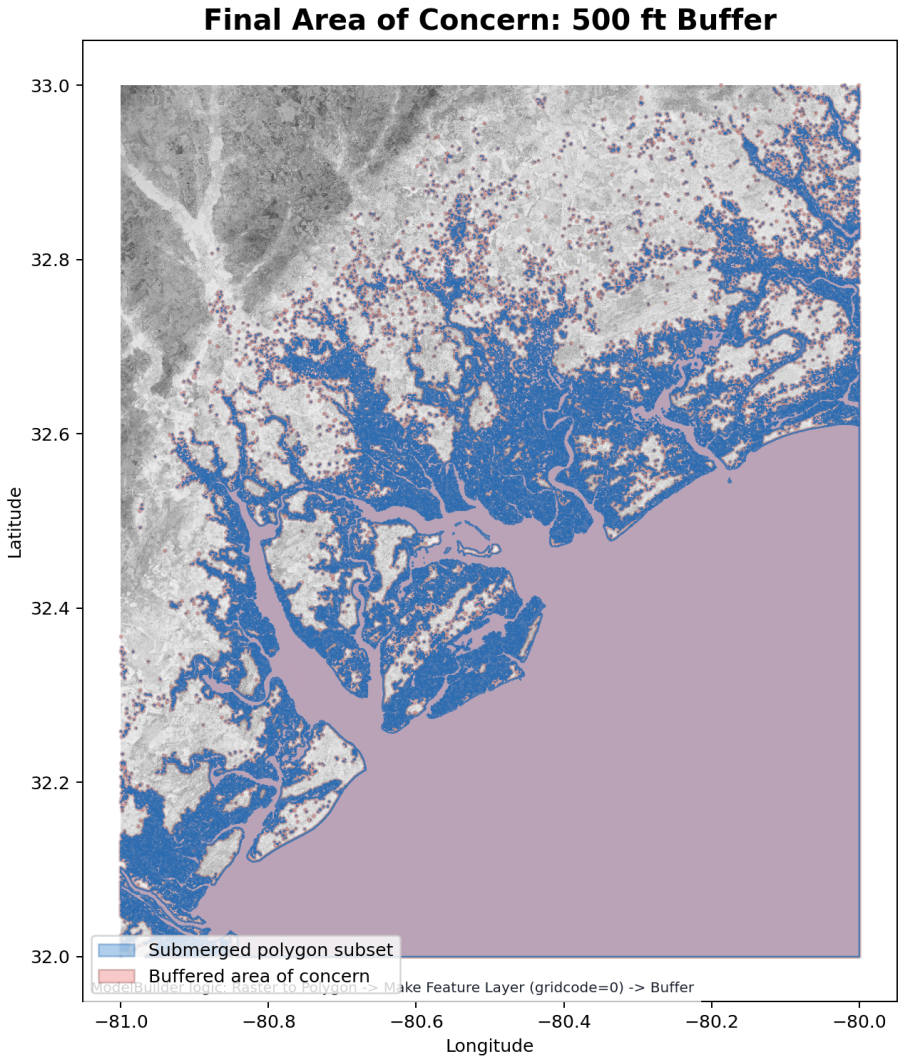
The Python run mirrors the ModelBuilder operation order. It does not replace the required ArcGIS toolbox submission; it verifies the same raster logic and records the expected outputs for review.

Sea Level Scenario Map



Orange cells represent the default 0-6 ft scenario band; blue cells are already at or below 0 ft. This is the visual counterpart to the 0/1 Raster Calculator output.

Final Area of Concern



The red overlay is the dissolved submerged polygon buffered by the default 500 ft area-of-concern parameter.

Technical Appendix

Output	Path
Scenario map	AIs/docs/unit07/unit07_sea_level_scenario_map.png
Area-of-concern map	AIs/docs/unit07/unit07_area_of_concern_map.png
Model steps CSV	AIs/docs/unit07/unit07_model_steps.csv
Scenario summary CSV	AIs/docs/unit07/unit07_scenario_summary.csv
Data manifest CSV	AIs/docs/unit07/unit07_data_manifest.csv
Inundation polygons	AIs/docs/unit07/unit07_inundation_polygons.geojson
Area-of-concern polygon	AIs/docs/unit07/unit07_area_of_concern.geojson

Traditional submission reminder: export screenshots of the ModelBuilder canvas, the tool dialog showing exposed parameters and default values, and the final output layer in the map. Submit those screenshots as the required PDF together with the toolbox file.